

ARUN KUMAR MOKASHI

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Summary

Computer Science student with experience in software development, backend systems, real-time applications, and automation. Built GitHub analysis and security tools, along with low-latency applications involving networking and real-time communication. Also worked on projects across web development, systems, and applied AI.

Education

B.E. in Computer Science

KLE Technological University

CGPA: 8.84/10

Aug 2023 - May 2027

Skills

Languages : Java, Python, JavaScript, SQL, HTML/CSS.
Frameworks & APIs : React, FastAPI, Flask, Streamlit, REST APIs, WebSockets, WebRTC.
Databases & Tools : MySQL, Git, GitHub, GitHub Actions, Docker, n8n, Linux, CI/CD, Unit Testing.
AI/ML : Machine Learning, Deep Learning, Reinforcement Learning, OpenCV, NLP, LangChain.
Core CS : Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks.

Experience

Software Development Intern – Arya Technologies | *Python, FastAPI, GitHub Actions* **Apr 2026 - Present**

- Built an Agile, microservices-based GitHub dashboard used by 1,000+ developers across 500+ repositories, centralizing CI/CD health, repository metadata, and dependency insights.
- Designed REST APIs and a MySQL-backed dependency-analysis module supporting 7 package ecosystems, parsing manifests and lockfiles to assess library security health and risk.
- Created a GitHub Actions security workflow to scan exposed credentials and syntax issues, block unsafe pull requests, and surface security artifacts as critical alerts on the dashboard UI in under 1 second.

Projects

VirtFrame (Wireless Touch Display) | *Android, Python, WebRTC, TCP/IP* **Mar 2026 - Jun 2026**

- Built a system that streams a Windows screen to an Android phone and maps mobile touch to PC controls.
- Implemented hardware-accelerated, low-latency WebRTC streaming with mss at 25-30 FPS for a responsive stream.
- Designed a TCP input protocol using `pynput` to translate six touch actions, including movement, clicks, and scrolling, into PC mouse controls with latency under 20 ms.

Saarthi: Mythology AI Chatbot | *Python, Flask, JS* **Oct 2025 - Nov 2025**

- Built a Flask web chatbot that answers Mahabharata queries and offers Krishna-inspired spiritual guidance.
- Developed 5 REST API endpoints to manage session creation, chat requests, and conversation cleanup, and implemented real-time streaming of AI responses to the web interface.
- Implemented UUID-based sessions and a sliding-window context manager that retains the system prompt and 10 recent messages, employing a RAG-lite approach with a JSON knowledge base for consistent responses.

Research Paper

Q-Learning Driven Real-Time Road Hazard Detection in Hilly Terrains | *YOLOv11* **Dec 2025**

- Published in Springer LNEE and presented at SGCNSP (2025), detailing a real-time system to find road hazards.
- Trained a YOLOv11 AI model on custom images to detect potholes and lane lines with 0.845 mAP, and used a Q-learning agent to smartly decide when an alert is dangerous enough to warn the driver.
- Integrated a Telegram bot using async Python to send instant hazard warning messages directly to the driver's phone.

Certifications & Achievements

- Solved 300+ problems on LeetCode, covering Data Structures and Algorithms.
- Won the Best Minor Project Award during the 6th semester.
- Leadership: Core Member of the college Programming Club, participating in competitive programming events.
- Certifications: OCI 2025 AI Foundations Associate; OCI 2025 Foundations Associate; Machine Learning Specialization, Stanford Online; NPTEL Elite in Machine Learning.